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RESEARCH ARTICLE

SCD-YOLO: Frequency-Domain Cleansing and Refinement for Lightweight Steel Surface Defect Detection

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ABSTRACT Surface defect detection underpins automated quality control in modern manufacturing, where inspection must be both accurate and lightweight to keep pace with continuous production lines. Detecting flaws on continuous metal sheets is difficult because the captured images are dominated by intense periodic background textures and contain low-contrast defects with weak gradients, both of which confound conventional spatial-domain feature extractors. We propose SCD-YOLO, an efficient end-to-end detector built on the YOLO26n baseline. We replace the C2PSA attention block at the backbone tail with the Orthogonal Spectral Decoupling Module (OSDM), which suppresses dominant background frequencies in an amplitude-only, sample-adaptive manner while preserving phase. Two Fractional-Order-Inspired Convolution Refinement (FOCR) modules are inserted at the P4 and P5 branches of the neck, applying a bounded fractional-order-inspired amplification that sharpens the weak gradients of low-contrast defects. The two modules follow a cleanse-then-refine principle, in which OSDM first reduces the dominant background and FOCR then amplifies the surviving defect signal. On the GC10-DET test set, SCD-YOLO achieves the highest mAP@0.5 of 68.8% and mAP@0.5:0.95 of 35.4% among all compared methods, improving the YOLO26n baseline by 3.2 and 1.9 percentage points, respectively, while using only 2.24M parameters and 5.1 GFLOPs, both below the baseline's 2.38M and 5.2 GFLOPs. On the NEU-DET test set, SCD-YOLO achieves 77.7% mAP@0.5 and the highest mAP@0.5:0.95 of 44.8%, improving the baseline by 0.9 and 0.6 percentage points, respectively, showing that the architectural benefit remains effective on a different steel surface defect distribution.

INDEX TERMS YOLO26, steel surface defect detection, frequency-domain feature processing, fractional-order calculus, lightweight object detection.

I. INTRODUCTION

Modern continuous production lines are moving toward Zero-Defect Manufacturing (ZDM) [1], where full in-line surface inspection supports closed-loop quality control on continuous metal sheets [2]. Such inspection must stay accurate while remaining light enough to match production throughput [3]. Deep learning, and the YOLO series in particular, has largely replaced manual inspection for this purpose. Yet deploying these models in real metallurgical environments exposes several practical weaknesses [4].

These weaknesses trace back to the harsh conditions of metal sheet production. Surface images suffer from two problems at once. Intense periodic textures, such as mechanical rolling marks, spread across the background, while the defects themselves, such as subtle creases, are faint and low in contrast [5]. This combination forces a trade-off. Small flaws lose their weak gradients during downsampling unless

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the input stays at high resolution, so resolutions such as 1024 × 1024 are often preferred [6]. Heavy models cope with such resolutions but cost too much computation for high-throughput inspection [7]. Lightweight models face a subtler problem. At high resolution the rolling marks generate strong, repetitive activations, and spatial attention such as C2PSA [8] weighs features by activation magnitude. The attention can then drift toward these dominant background responses and away from the faint defects it should localize.

This study proposes SCD-YOLO (Spectral Cleansing and Decoupling YOLO) to address this problem. Since spatial attention can be misled when high-amplitude periodic noise dominates the feature space, we replace the C2PSA block in the YOLO26n backbone with the Orthogonal Spectral Decoupling Module (OSDM), moving the final-stage refinement from spatial attention to frequency-domain decoupling. Frequency-domain processing offers a complementary way to handle periodic backgrounds. Rolling marks tend to produce concentrated high-amplitude spectral responses [9], whereas non-periodic defect signals are generally distributed more broadly. This difference represents a data-dependent spectral tendency rather than a strict separation between background and defect components. Operating within this cleansed feature space, we further insert two Fractional-Order-Inspired Convolution Refinement (FOCR) modules into the bottom-up fusion path at the P4 and P5 branches of the neck. Inspired by the frequency response of fractional differentiation, FOCR employs a bounded spectral amplify-ication profile to strengthen weak defect gradients after OSDM has reduced dominant background interference. Together the two modules implement a cleanse-then-refine principle, which we validate through ablation in Section IV.

The main contributions of this work are as follows:

1) Proposed SCD-YOLO, an efficient end-to-end detector for industrial surface defect detection. It replaces the C2PSA block in YOLO26n with two complementary spectral modules, improving detection accuracy while retaining a compact parameter count and nominal computational complexity.
2) Designed the Orthogonal Spectral Decoupling Module (OSDM), placed at the backbone tail, which suppresses dominant background frequencies in a sample-adaptive, amplitude-only manner while preserving phase, reduce background interference at the deepest semantic stage.
3) Introduced the Fractional-Order-Inspired Convolution Refinement (FOCR) module into the bottom-up fusion path at the P4 and P5 branches. Operating on features already cleansed by OSDM, FOCR applies a bounded fractional-order-inspired amplification that enhances the weak gradients of low-contrast defects.
4) Evaluated SCD-YOLO on held-out test sets of GC10-DET and NEU-DET. On GC10-DET, it achieves 68.8% mAP@0.5 and 35.4% mAP@0.5:0.95, improving the YOLO26n baseline by 3.2 and 1.9 percentage points, respectively, while retaining only 2.24M parameters and 5.1 GFLOPs. Independent evaluation on NEU-DET further confirms improvements over the YOLO26n baseline under a different steel surface defect distribution.

II. RELATED WORK

A. DEEP LEARNING FOR SURFACE DEFECT DETECTION

Deep learning–based surface defect detection has progressed through several stages. Early work was dominated by two-stage detectors built on the Faster R-CNN paradigm. Dong et al. [10] proposed PGA-Net, which combines pyramid feature fusion with global context attention to localize defects on metallic surfaces. Wang et al. [11] extended Cascade R-CNN with deformable convolutions to handle severe defect deformations. These methods reach strong accuracy, but their dense anchor generation and multi-stage inference are hard to reconcile with the throughput of modern continuous production lines.

Single-stage detectors, especially the YOLO series, closed much of this speed gap and became the prevailing choice. Zhou et al. [12] applied YOLOv5 with decoupled heads to steel surface defects, balancing classification and regression. Anchor-free designs in YOLOv8 and YOLO11 [13] then simplified the pipeline by removing the hand-designed anchors that had required dataset-specific tuning. YOLO26n [14] went further, dropping non-maximum suppression and other empirical priors to enable end-to-end inference. Like other purely spatial-domain detectors, however, YOLO26n has no explicit mechanism to filter out unconstrained background noise, which can limit its robustness under heavy periodic textures such as rolling marks.

Vision Transformers have also been explored for their global receptive fields. Mia and Li [15] used a Swin Transformer to capture long-range dependencies in surface images, and Li et al. [16] adapted RT-DETR with global context modeling to suppress localized background interference. Their representational strength comes at a cost: the FLOPs and memory footprint of Transformer detectors remain high, which keeps lightweight CNNs competitive when resources are tight.

Within the lightweight CNN line, attention mechanisms and multi-scale feature fusion are commonly incorporated into YOLO variants to improve feature representation under complex industrial conditions. Jin et al. [17] integrated structural spatial attention to emphasize irregular defect contours, and Kang et al. [18] combined spatial pyramid pooling with CBAM to refine multi-scale features. Kothapalli et al. [19] proposed C3SE-YOLO by integrating C3SE blocks, BiFPN, and SE attention into YOLOv5 to strengthen feature extraction and multi-scale feature aggregation. Ancha et al. [20] proposed CSPGhost-YOLO by combining GhostC3 and parameter-free SimAM attention with YOLOv5, resulting in a lightweight architecture for real-time PCB defect detection and edge deployment, while their subsequent

TRSBi-YOLO [21] combined C3TR, SimAM, and BiFPN to improve multi-scale representation within a compact detector. From a deployment perspective, Bakirci [22] systematically evaluated lightweight YOLO variants on a resource-constrained Jetson Xavier NX platform in terms of detection accuracy, inference latency, GPU utilization, and power consumption. The results show that practical lightweight detection must balance accuracy, latency, and energy efficiency rather than rely solely on parameter count or FLOPs.

These studies demonstrate the effectiveness of lightweight spatial, channel-attention, and multi-scale feature-enhancement strategies. However, they primarily refine features in the spatial or channel domain and do not explicitly model the dominant periodic interference that appears in the frequency spectrum of steel surfaces.

B. ATTENTION AND FREQUENCY-DOMAIN METHODS FOR FEATURE ENHANCEMENT

Spatial and channel attention mechanisms are widely used in defect detection to offset the locality of standard convolutions. Modules such as CBAM and C2PSA reweight feature maps to emphasize task-relevant regions. Liu et al. [23] used a dual-attention module to highlight defect contours against uniform backgrounds, and deformable variants have captured irregular defect geometries [24]. These methods share a structural weakness rooted in how they assign importance. Attention is driven by activation magnitude, it responds most strongly where contrast and activation are high. When intense rolling marks fill the background with sharp, repetitive gradients, the strongest activations come from the background itself, and the attention concentrates there instead of on the faint defects that matter.

A complementary line of work moves the processing into the frequency domain. Qin et al. [25] showed in FcaNet that global average pooling is the lowest-frequency component of the 2D DCT, and that adding further DCT components yields richer channel descriptors. Rao et al. [26] proposed GFNet, which learns long-range dependencies through global filters applied via FFT, demonstrating that frequency-domain filtering can replace spatial operators in vision tasks. The frequency view fits the present problem well. Quasi-periodic rolling marks tend to produce concentrated high-amplitude spectral responses, whereas non-periodic defects are generally more broadly distributed, although spectral overlap may still occur.

Existing spectral filters share a limitation. Methods such as GFNet learn one global set of frequency weights and apply it unchanged to every input. Yet the dominant frequency and amplitude of rolling marks shift across samples, rolling conditions, and steel grades, and a globally shared filter cannot follow this per-sample variation. A spectral module that adapts its suppression to each input therefore remains an open need for industrial inspection, and this is the gap our OSDM addresses.

C. FRACTIONAL-ORDER CALCULUS IN FEATURE ENHANCEMENT

Once periodic background noise is suppressed in the frequency domain, a second problem remains: amplifying the faint signals of low-contrast defects such as shallow scratches and weak creases. In conventional CNNs, edge enhancement relies on standard spatial convolutions, which approximate integer-order differentiation. These operators follow a power-law frequency response of the form $|\omega|^n$ with integer order n. First-order operators amplify high frequencies linearly with ω and boost real edges and high-frequency noise alike, while low-pass smoothing does the reverse, attenuating noise but blurring subtle gradients. An integer-order formulation cannot easily escape this trade-off between noise suppression and edge preservation [27].

Fractional-order calculus (FOC) interpolates between these extremes. Pu et al. [28] applied fractional-order differentiation to multiscale texture enhancement and showed that the frequency response of a fractional derivative of order v takes the form $|\omega|^v$, where v can be non-integer between 0 and 1. In this range the response amplifies mid-to-high frequencies more gently than integer-order differentiation and attenuates low frequencies less aggressively. The fractional exponent thus acts as a continuous control between low-pass smoothing at $v = 0$ and first-order differentiation at $v = 1$, which makes FOC well suited to lifting the subtle gradients of low-contrast defects without over-amplifying high-frequency noise.

Bringing FOC into end-to-end networks has been less straightforward. As Joshi et al. [29] review, traditional FOC in image processing applies fractional-order masks as a static, offline preprocessing step, which rules out task-driven optimization. More recent work embeds fractional-order layers inside CNNs [30], but these rely on dense, large-kernel spatial approximations that add appreciable computation. A lightweight, residual-based fractional-order refinement module that complements upstream spectral decoupling therefore remains an open practical problem.

III. METHODS

A. OVERALL ARCHITECTURE OF SCD-YOLO

SCD-YOLO builds on the YOLO26n baseline [14]. The base architecture extracts features through a C3K2 backbone and an SPPF multi-scale pooling layer [31], fuses semantics through a bidirectional PANet neck [32], and predicts boxes with an anchor-free, NMS-free end-to-end head. This pipeline is efficient, but it has no explicit mechanism for handling severe periodic noise or low-contrast gradients.

SCD-YOLO modifies the topology in two places. First, it replaces the C2PSA attention block at the backbone tail with the OSDM, moving the final-stage refinement from spatial attention to frequency-domain decoupling. Second, it inserts two FOCR modules into the bottom-up path of the neck, placed immediately after the C3K2 fusion blocks at the P4 and P5 branches. All other components, including the

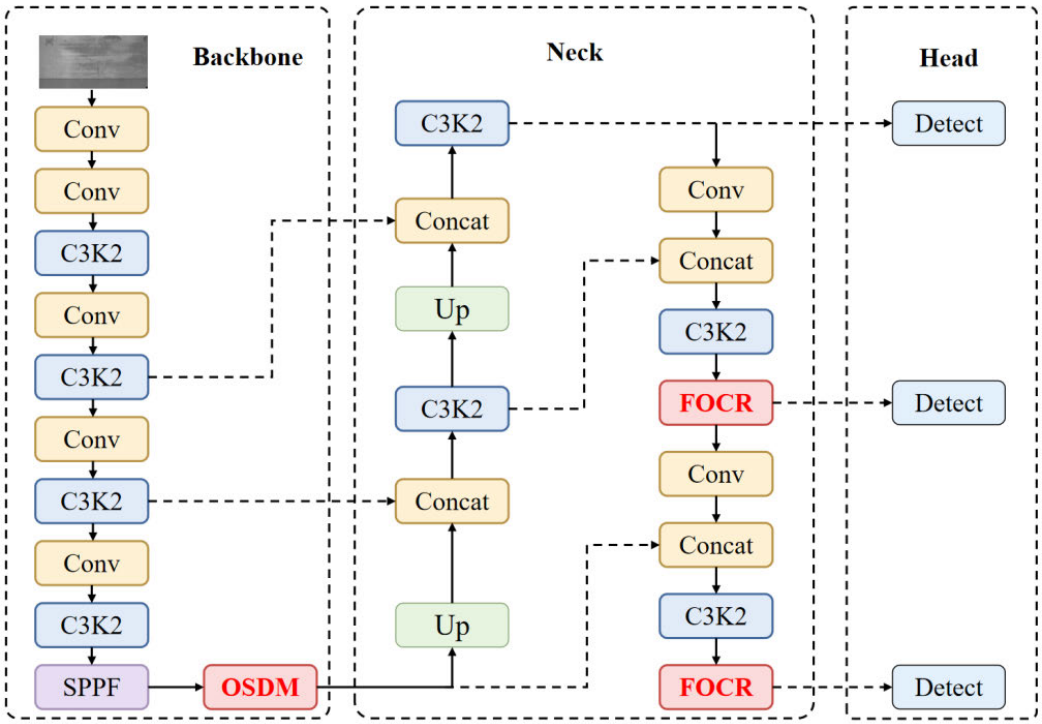


FIGURE 1. SCD-YOLO network architecture.

C3K2 backbone blocks, the SPPF layer, the FPN top-down path, and the detection head, stay unchanged.

As Figure 1 shows, these two changes realize a cleanse-then-refine design. OSDM first suppresses periodic background interference at the deepest semantic stage (P5), and the cleansed high-level features propagate down through the FPN path [33]. The FOCR modules then operate on the bottom-up fused features at the P4 and P5 branches, which lie closest to the OSDM-cleansed output and therefore carry the least residual background. Applying fractional-order-inspired spectral refinement here strengthens the surviving weak defect responses. The high-resolution P3 branch is left unchanged: it sits farthest from OSDM and retains the most background texture, so amplifying it would boost that residual noise faster than the defect signal, an effect we confirm in the placement study of Section IV.

B. ORTHOGONAL SPECTRAL DECOUPLING MODULE (OSDM)

The deep semantic features at P5 carry two kinds of high-amplitude content: narrow-band peaks from the periodic rolling marks, and broader high-amplitude regions tied to the background. Periodic background structures tend to produce concentrated high-amplitude spectral responses, whereas irregular defect information is generally distributed more broadly. OSDM exploits this data-dependent tendency by treating the Top-k high-amplitude components as dominant spectral candidates and softly down-weighting them while retaining the original representation through a residual connection. The module works entirely on the orthonormal Fourier basis [34], where modifying one frequency component leaves the others algebraically unchanged, so each high-amplitude position can be suppressed on its own. Unlike spectral networks that apply one input-agnostic global filter, OSDM suppresses amplitudes in a sample-adaptive way, following the spectral signature of each input.

Figure 2 shows the architecture of OSDM. Given an input feature map $X \in \mathbb{R}^{C\times H\times W}$ from the P5 stage, OSDM applies the 2D real-valued Fast Fourier Transform ($\mathcal{F}_{2D}$) and splits the complex spectrum into amplitude and phase:

$$\mathcal{Z} = \mathcal{F}_{2D}(X) \tag{1}$$

$$\mathcal{M} = |\mathcal{Z}| \tag{2}$$

$$\Phi = \angle\mathcal{Z} \tag{3}$$

where $\mathcal{M}, \Phi \in \mathbb{R}^{C\times H\times(W/2+1)}$. The conjugate symmetry of the rFFT halves the spectral storage without losing information. Background interference concentrates in the high-amplitude regions of $\mathcal{M}$, whereas the phase Φ encodes the spatial layout of features [35]. OSDM therefore modifies only $\mathcal{M}$ and leaves Φ untouched, preserving the spatial geometry of the defect signals through the suppression step.

The suppression strength need adapt to the varying rolling-mark patterns of each input. Therefore, OSDM derives a sample-dependent through a lightweight channel descriptor:

$$\mathbf{w} = \sigma\left(W_2 \cdot \mathrm{LeakyReLU}\left(W_1 \cdot \mathrm{GAP}(X)\right)\right), \mathbf{w} \in \mathbb{R}^C \tag{4}$$

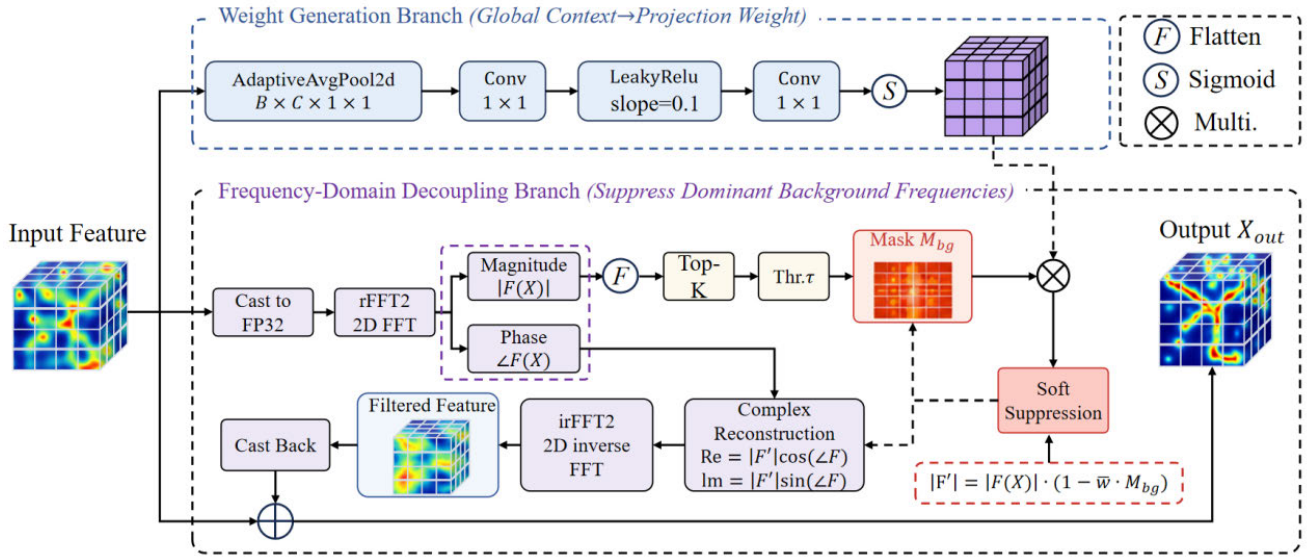


FIGURE 2. OSDM architecture.

where GAP denotes global average pooling, $W_1 \in \mathbb{R}^{c_{\text{inter}} \times C}$ and $W_2 \in \mathbb{R}^{C \times c_{\text{inter}}}$ are 1×1 convolutions with hidden channels $c_{\text{inter}} = \max(C/4, 16)$, and σ is the sigmoid activation. The C-dimensional output $\mathbf{w}$ is avera-ged into a sample-level suppression strength shared across channels, while the spectral mask remains sample- and channel-dependent:

$$\alpha = \frac{1}{C} \sum_{i=1}^{C} w_i \in (0, 1) \tag{5}$$

The sigmoid constraint keeps α below 1, dominant frequency components are attenuated rather than explicitly removed. This continuous suppression helps avoid abrupt feature distortion and supports stable end-to-end optimization.

The set of frequencies to suppress is selected independently for each sample and channel. For each sample-channel pair, OSDM ranks the flattened spectral amplitudes and uses the k-th largest value as the threshold for identifying dominant spectral candidates. k is a fixed integer that controls the number of highest-amplitude spectral positions considered for suppression; it is not scaled with the channel number or feature-map size. In the final architecture, OSDM is applied only at the P5 backbone tail, and $k = 3$ is used throughout the experiments. Let $\tau_{b,c}$ denote the k-th largest value in the flattened amplitude map:

$$\tau_{b,c} = \text{TopK}\left(\text{vec}(\mathcal{M}_{b,c,:,:}), k\right)_k \tag{6}$$

The background mask is the indicator function:

$$\mathcal{B}_{b,c,h,w} = 1\left[\mathcal{M}_{b,c,h,w} \geq \tau_{b,c}\right] \tag{7}$$

Because $\tau_{b,c}$ is computed from each sample's own amplitude distribution, the mask $\mathcal{B}$ adapts to the spectral signature of every input, complementing the input-dependent strength α described above.

Suppression is applied softly to the amplitude only, and the spectrum is reassembled with the original phase. The filtered amplitude $\mathcal{M}'$, the reconstructed spectrum, and the spatial-domain output are given by:

$$\mathcal{M}' = \mathcal{M} \odot (1 - \alpha \cdot \mathcal{B}) \tag{8}$$

$$\mathcal{Z}' = \mathcal{M}' \odot e^{j\Phi} \tag{9}$$

$$Y = \mathcal{F}_{2D}^{-1}\left(\mathcal{Z}'\right) + X \tag{10}$$

The element-wise suppression factor $1 - \alpha\mathcal{B}$ scales the identified high-amplitude positions by $(1 - \alpha) \in (0, 1)$ and leaves all other frequencies unchanged. The residual connection in the spatial-domain output makes OSDM's contribution additive rather than substitutive, which stabilizes training and lets the module be integrated into the YOLO26n architecture without altering downstream feature dimensions. The orthonormal FFT used throughout preserves the ℓ_2 energy between spatial and spectral domains.

OSDM introduces only a small learnable-parameter overhead. The channel descriptor consists of two 1×1 convolutions whose parameter count is approximately $C \cdot c_{\text{inter}} + c_{\text{inter}} \cdot C$. In the YOLO26n configuration used here, the P5 stage has $C = 256$ and $c_{\text{inter}} = \max(C/4, 16) = 64$, giving about 33K parameters for the descriptor. The 2D rFFT, top-k selection, masking, and inverse FFT contribute little to the nominal GFLOP count compared with the convolutional layers of the backbone, but their practical runtime cost is not fully captured by conventional GFLOP estimates. Because OSDM replaces the C2PSA attention block at the backbone tail, the net parameter count of the P5 stage drops by roughly 0.22M, as confirmed in the ablation of Section IV. To improve numerical stability under low-precision inference, OSDM internally upcasts the feature map to FP32 for the spectral operations and casts the

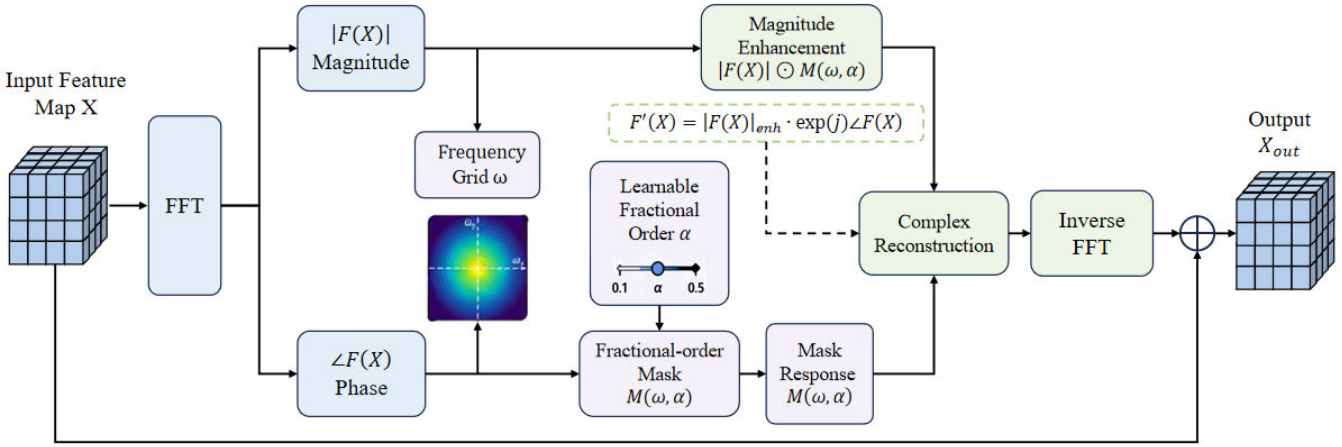


FIGURE 3. FOCR architecture.

reconstructed output back to the native dtype before residual addition. Therefore, although OSDM remains compact in terms of learnable parameters and nominal computation, its actual latency and memory overhead are hardware dependent.

C. FRACTIONAL-ORDER-INSPIRED CONVOLUTION REFINEMENT (FOCR)

After OSDM reduces periodic background interference at P5, the cleaned high-level features propagate down through the FPN top-down path and are then fused back up along the PANet bottom-up path. At this stage, the multi-scale features carry less background interference, but the gradients of low-contrast defects such as creases remain weak. Section II reviewed how fractional-order calculus provides a principled way to amplify mid-to-high frequency components, where these weak gradients reside, without indiscriminately boosting noise. We insert two FOCR modules into the bottom-up path immediately after the C3K2 fusion blocks at the P4 and P5 scales, each operating independently on its own feature scale.

Figure 3 shows the overall architecture of FOCR. Given the fused feature map $X \in \mathbb{R}^{C\times H\times W}$ at either P4 or P5, FOCR follows the same spectral convention as OSDM. It applies the 2D real-valued FFT to obtain$\mathcal{Z}$ decomposes it into amplitude $\mathcal{M}$ and phase Φ, modifies only $\mathcal{M}$, and reassembles the spectrum with the original phase. The two modules thus share a common spectral framework but pursue opposite goals. OSDM suppresses dominant noise peaks, while FOCR amplifies the broadband mid-frequency content where weak edges reside.

The amplification profile is governed by a single learnable fractional order. We denote the raw parameter $\hat{\alpha} \in \mathbb{R}$ and apply a clamp to maintain training stability:

$$\alpha = \text{clamp}(\hat{\alpha}, 0.1, 0.5) \tag{11}$$

The clamp range [0.1, 0.5] keeps the response in a regime where moderate edge amplification dominates over high-frequency noise boosting. Unlike OSDM, whose suppression strength is computed per sample to track the highly variable spectral signature of rolling marks, α in FOCR is a single learnable scalar shared across all inputs. Placed downstream of OSDM, FOCR learns a global amplification profile that generalizes across defect samples rather than reacting to individual ones.

Standard fractional differentiation has a frequency response proportional to $|\omega|^{\alpha}$, which grows without bound as $\omega \to \infty$ and would amplify residual high-frequency noise as aggressively as real edges. FOCR should therefore be understood as a fractional-order-inspired spectral refinement module rather than a direct implementation of a classical fractional derivative. Specifically, it retains the power-law term ω^{α} associated with fractional-order differentiation and embeds it into a bounded and learnable frequency response. We first construct the radial frequency grid:

$$\omega_{u,v} = \sqrt{u^2 + v^2 + \varepsilon} \tag{12}$$

$$u = \frac{2\pi k_u}{H} \tag{13}$$

$$v = \frac{2\pi k_v}{W} \tag{14}$$

where k_u and k_v are the discrete spatial frequency indices, the factor 2π converts normalized frequencies into angular frequencies, and $\varepsilon = 10^{-8}$ avoids singularities at the origin. The resulting fractional-order-inspired amplification mask is defined as:

$$m(\omega) = 1 + \frac{\omega^{\alpha}}{1 + \omega^{\alpha}} \tag{15}$$

This mask has three properties that motivate its design. When $\omega \to 0, \omega^{\alpha} \to 0$ and $m \to 1$, the response to DC and low-frequency components remains close to identity,

thereby preserving the global structural and semantic information. As ω grows, the fraction saturates and caps the maximum amplification at 2.0 in the limit. The transition between 1.0 and 2.0 is shaped by α. This bounded form retains the power-law frequency-shaping characteristic inspired by fractional-order differentiation while avoiding the unbounded high-frequency amplification that could destabilize end-to-end training in the presence of residual noise.

Amplification is applied element-wise to the amplitude, and the spectrum is reassembled:

$$\mathcal{M}' = \mathcal{M} \odot m(\omega) \tag{16}$$

$$\mathcal{Z}' = \mathcal{M}' \odot e^{j\Phi} \tag{17}$$

The spatial-domain output is recovered by the inverse FFT, passed through a 1×1 convolution, Batch Normalization, and SiLU activation, and finally combined with the input through a residual connection:

$$\tilde{X} = \mathcal{F}_{2D}^{-1}(\mathcal{Z}') \tag{18}$$

$$Y = \mathrm{SiLU}(\mathrm{BN}(W_c\tilde{X})) + X \tag{19}$$

where $W_c \in \mathbb{R}^{C\times C}$ is a 1×1 convolutional kernel. The residual connection makes FOCR's contribution additive rather than substitutive on the fused PANet features. In the YOLO26n configuration used here, the P4 and P5 feature dimensions are $C = 128$ and $C = 256$ respectively. The two FOCR instances add approximately 0.08M parameters in total. Similar to OSDM, the rFFT, spectral modulation, and inverse rFFT operations are not fully characterized by conventional GFLOP estimates. FOCR also performs the spectral operations in FP32 before casting the reconstructed feature map back to the native dtype.

IV. EXPERIMENTS

A. DATASETS

Two public surface defect datasets are employed to evaluate SCD-YOLO: GC10-DET [36] serves as the primary benchmark on cold-rolled steel, while NEU-DET [37] serves as an independent external benchmark on hot-rolled steel with a different image resolution, defect taxonomy, and background distribution, thereby providing a independent cross-dataset evaluation. Their image resolutions, train/validation/test splits, and dataset characteristics are described below, while the class-wise image and bounding-box annotation distributions are summarized in Table 1.

1) THE GC10-DET DATASET

The GC10-DET dataset is a public benchmark for steel surface defect detection, which is collected from cold-rolled steel production lines. It contains 2,280 grayscale images at an original resolution of 2048 × 1000 pixels, covering ten defect categories: punching, weldline, crescent gap, water spot, oil spot, silk spot, inclusion, rolled pit, crease, and waist folding.

This dataset has two main challenges. The original resolution is high, so small defects such as faint creases and inclusions occupy very few pixels, and detectors must recover their weak gradient signals across multiple scales. The defects also sit against the heavy background textures of cold-rolled steel, including rolling marks, oil residues, and periodic surface patterns. These textures mask defect features in the spatial domain.

We partition the dataset into training, validation, and testing sets in an 8:1:1 ratio, yielding 1,824, 228, and 228 images respectively.

2) THE NEU-DET DATASET

The NEU-DET dataset [37] is a public benchmark for hot-rolled steel strip surface defect detection. The dataset contains 1,800 grayscale images at an original resolution of 200 × 200 pixels, covering six defect categories: crazing, inclusion, patches, pitted surface, rolled-in scale, and scratches.

NEU-DET differs from GC10-DET in two important respects. First, the lower original resolution and smaller image size place stricter demands on small-object localization. Second, the surface texture of hot-rolled steel exhibits different periodic patterns from those of cold-rolled steel, providing a complementary benchmark for assessing whether the proposed architecture remains effective under a different industrial surface distribution when trained independently on that dataset. We adopt the same 8:1:1 split as GC10-DET, yielding 1,440 training, 180 validation, and 180 testing images. Its class-wise image and annotation statistics are also included in Table 1.

Representative unaugmented samples of all defect categories in GC10-DET and NEU-DET are shown in Figure 4.

TABLE 1. Dataset composition and annotation statistics.

Dataset	GC10-DET	NEU-DET
Original Resolution	2048×1000	200×200
Train/Val/Test	1824/228/228	1440/180/180
Total Annotations	3541	4108
Class-wise Images / Annotations	C1: 324/324; C2: 505/506; C3: 262/263; C4: 308/352; C5: 250/569; C6: 734/884; C7: 198/344; C8: 44/83; C9: 53/74; C10: 139/142	C1: 300/689; C2: 374/981; C3: 340/875; C4: 301/430; C5: 298/608; C6: 300/525

Each class-wise entry denotes the number of images containing the corresponding class / the number of bounding-box annotations. For GC10-DET, C1–C10 follow the class order of punching, weldline, crescent gap, water spot, oil spot, silk spot, inclusion, rolled pit, crease, and waist folding. For NEU-DET, C1–C6 correspond to crazing, inclusion, patches, pitted surface, rolled-in scale, and scratches.

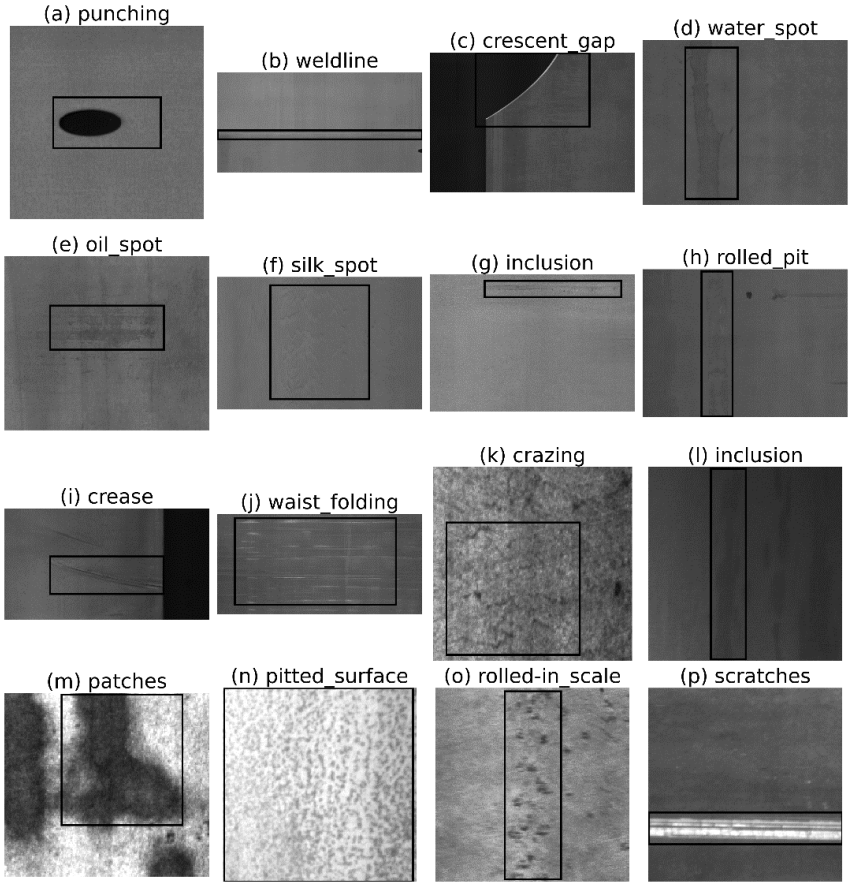


FIGURE 4. Representative defect samples from the two datasets. (a)–(j) GC10-DET: punching, weldline, crescent gap, water spot, oil spot, silk spot, inclusion, rolled pit, crease, and waist folding; (k)–(p) NEU-DET: crazing, inclusion, patches, pitted surface, rolled-in scale, and scratches. All patches are cropped from the original images without data augmentation, and the boxes indicate the ground-truth defect regions.

TABLE 2. Basic experimental environment configuration.

Category	Item	Specification
Hardware	GPU	RTX 4090 (24 GB)
	CPU	Intel Core i9-14900K
	RAM	64 GB
Software	Operating System	Windows 11
	Python	3.10
	PyTorch	2.10
	CUDA	12.6
Training Settings	GC10-DET Input image resolution	1024×1024
	NEU-DET Input image resolution	640×640
	Optimizer	MuSGD
	Initial Learning Rate	0.01
	Momentum	0.937
	Weight Decay	5×10^{-4}
	Total Epochs	300
	Batch Size	16

TABLE 3. Data-augmentation settings used in the experiments.

Category	Augmentation / Parameter	Value
Photometric	HSV hue gain	0.015
	HSV saturation gain	0.70
	HSV value gain	0.40
Geometric	Translation	0.10
	Scaling	0.50
	Horizontal flip probability	0.50
Multi-image	Mosaic probability	1.00
Schedule	Close Mosaic	Final 10 epochs
Disabled	Rotation	0.0
	Shear	0.0
	Perspective	0.0
	Vertical flip	0.0
	MixUp	0.0
	CutMix	0.0
	Copy-paste	0.0
	BGR channel swapping	0.0

B. IMPLEMENTATION DETAILS

1) EXPERIMENTAL SETUP

For controlled comparison, all models are trained from scratch without COCO or ImageNet pretraining and use the same dataset splits, input resolutions, training duration, and evaluation protocol. Models implemented within

the same framework also share the optimizer and data-augmentation settings wherever applicable. The hardware, software, and tr-aining configurations are summarized in Table 2. All architecture-selection experiments, including module ablation, hyperparameter sensitivity analysis, and placement studies, are conducted exclusively on the GC10-DET validation set. The test sets are used only for final comparative evaluation and are not involved in architecture or hyperparameter selection.

The training-time data-augmentation configuration is summarized in Table 3. The pipeline includes HSV-based photometric augmentation, translation, scaling, horizontal flipping, and Mosaic augmentation, with Mosaic disabled during the final 10 training epochs.

Representative examples of the adopted augmentation strategies are shown in Figure 5. Panels (a) and (d) present original training samples, panels (b) and (e) illustrate the single-image photometric and geometric transformations, and panels (c) and (f) show Mosaic augmentation on GC10-DET and NEU-DET, respectively. Mosaic is temporarily disabled only when generating panels (b) and (e) to isolate the single-image transformations for visualization.

To assess run-to-run variability, the core experiments are repeated with three random seeds (0, 1, and 2): four configurations (YOLO26n, OSDM-only, FOCR-only, and SCD-YOLO) on GC10-DET and YOLO26n/SCD-YOLO on NEU-DET. Results are reported as mean $\pm$ standard deviation on the corresponding validation sets. Runtime is measured for the four GC10-DET configurations over five validation runs on an RTX 4090 using 1024×1024 inputs, batch size 16, and half-precision inference, with latency reported as mean $\pm$ standard deviation and peak allocated GPU memory.

2) EVALUATION METRICS

We adopt the standard COCO evaluation protocol [38] to report detection performance. mAP@0.5 measures the mean Average Precision computed at the IoU threshold of 0.5, evaluating whether the model can localize defects with reasonable overlap with the ground truth. mAP@0.5:0.95 averages the mAP over ten IoU thresholds from 0.5 to 0.95 in steps of 0.05, providing a stricter measure of localization precision. We additionally report parameter count, GFLOPs, measured inference latency, and peak GPU memory to evaluate computational efficiency.

C. COMPARISON WITH STATE-OF-THE-ART METHODS

Seven detectors are included for comparison across the two steel surface defect datasets: Faster R-CNN with a ResNet50-FPN backbone, YOLOv5n, YOLOv8n, YOLO11n, the YOLO26n baseline, RT-DETR-L, and DEFECT-YOLO. The YOLO variants and RT-DETR-L are implemented in Ultralytics, Faster R-CNN in MMDetection, and DEFECT-YOLO using its official implementation. All models are trained from scratch using the predefined dataset splits and evaluated on the corresponding test sets. For models implemented in different frameworks, architecture-compatible optimization settings are retained while keeping the dataset split, input resolution, training duration, and evaluation protocol aligned.

Table 4 reports the GC10-DET test results. SCD-YOLO achieves the highest mAP@0.5 and mAP@0.5:0.95, reaching 68.8% and 35.4%, respectively, which are 3.2 and 1.9 percentage points higher than YOLO26n. These gains are obtained with substantially fewer parameters than RT-DETR-L and Faster R-CNN.

Beyond Table 4, the class-wise metrics reveal clear differences across defect categories. SCD-YOLO performs strongly on weldline, achieving 80.4% precision, 100.0% recall, 89.1% F1-score, and 97.5% AP@0.5. In contrast, rolled_pit remains the most challenging category, with 39.6% precision, 25.0% recall, 30.7% F1-score, and 36.9% AP@0.5, although its AP@0.5 increases substantially from 1.4% with YOLO26n. Inclusion and crease also remain difficult, with AP@0.5 values of 43.1% and 48.5%, respectively. These results highlight both the class-specific strengths and the remaining weaknesses of SCD-YOLO.

To assess the cross-dataset robustness of the proposed architecture, we independently retrain and evaluate all methods on NEU-DET at 640×640. Compared with GC10-DET, NEU-DET differs substantially in image resolution, defect categories, background texture, and steel-production domain, and therefore serves as an external cross-dataset benchmark. We restrict this comparison to the YOLO-series detectors and RT-DETR-L, since the GC10-DET results already establish the standing of Faster R-CNN and DEFECT-YOLO relative to SCD-YOLO. The test-set results are reported in Table 5.

SCD-YOLO improves the YOLO26n baseline on both metrics, increasing mAP@0.5 from 76.8% to 77.7% and mAP@0.5:0.95 from 44.2% to 44.8%. SCD-YOLO achieves the highest mAP@0.5:0.95 at 44.8%. Relative to YOLO26n, the gains are 0.9 and 0.6 percentage points on NEU-DET, compared with 3.2 and 1.9 percentage points on GC10-DET. This difference indicates that the relative benefit under loose and strict IoU criteria is dataset-dependent. Differences in image resolution, defect scale, and background texture may contribute to this variation, although the present experiments do not isolate a single causal factor. The improvement over YOLO26n on both datasets indicates that the benefit of the proposed spectral modules is not limited to GC10-DET.

D. ABLATION STUDIES

To examine the contribution of each module, we conduct ablation experiments on GC10-DET with four configurations: the YOLO26n baseline, baseline with OSDM only, baseline with FOCR only, and the full SCD-YOLO (baseline with both OSDM and FOCR). All configurations are trained under identical settings. Results are reported in Table 6.

Adding OSDM to the baseline improves mAP@0.5 from 71.1% to 74.1% (+3.0 percentage points) and mAP@0.5:0.95 from 37.8% to 38.2% (+0.4 percentage points). The accuracy

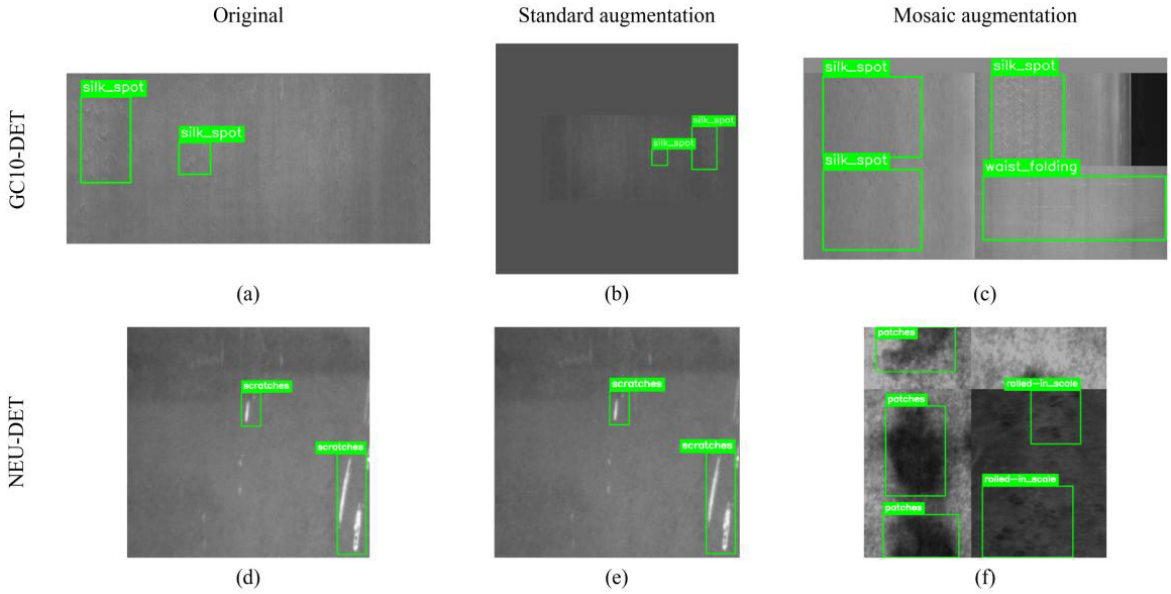


FIGURE 5. Representative training-time augmentation examples on GC10-DET and NEU-DET.

TABLE 4. Performance comparison with SOTA methods on GC10-DET test set.

Model	mAP@0.5 (%)	mAP@0.5:0.95 (%)	Params (M)	GFLOPs
Faster R-CNN	64.1	30.5	41.39	113.0
YOLOv5n	59.0	31.1	2.50	7.1
YOLOv8n	58.9	30.4	3.01	8.1
YOLO11n	56.7	29.2	2.58	6.3
RT-DETR-l	65.5	32.6	32.00	103.5
DEFECT-YOLO	65.1	34.2	3.29	7.9
YOLO26n (Baseline)	65.6	33.5	2.38	5.2
SCD-YOLO (Ours)	68.8	35.4	2.24	5.1

TABLE 5. Performance comparison on NEU-DET test set.

Model	mAP@0.5 (%)	mAP@0.5:0.95 (%)	Params (M)	GFLOPs
YOLOv5n	76.8	44.3	2.50	7.1
YOLOv8n	77.0	44.4	3.01	8.1
YOLO11n	78.2	44.7	2.58	6.3
RT-DETR-l	69.9	39.7	32.00	103.5
YOLO26n (Baseline)	76.8	44.2	2.38	5.2
SCD-YOLO (Ours)	77.7	44.8	2.24	5.1

TABLE 6. Ablation study of the proposed spectral modules on GC10-DET validation set.

Model	OSDM	FOCR	mAP@0.5 (%)	mAP@0.5:0.95 (%)	Params (M)	GFLOPs	Inference Latency (ms)	Peak Memory (MB)
YOLO26n (Baseline)	—	—	71.1	37.8	2.38	5.2	2.04 ± 0.35	299.87
+ OSDM	√	—	74.1	38.2	2.16	5.0	2.18 ± 0.37	299.46
+ FOCR	—	√	71.6	36.9	2.46	5.3	2.33 ± 0.77	316.18
SCD-YOLO (Ours)	√	√	75.3	38.7	2.24	5.1	2.27 ± 0.25	315.77

Latency is reported as mean ± standard deviation over five validation runs using batch size 16, 1024 × 1024 input, and half-precision inference on an NVIDIA RTX 4090; peak memory denotes the maximum allocated GPU memory.

gain is accompanied by a reduction in model complexity: the parameter count decreases from 2.38M to 2.16M and computation from 5.2 to 5.0 GFLOPs, since OSDM replaces the heavier C2PSA attention block at the backbone tail of YOLO26n. In the present YOLO26n configuration, replacing C2PSA with OSDM provides a more favorable parameter–accuracy trade-off.more favorable parameter–accuracy trade-off.

TABLE 7. Multi-seed robustness analysis on the GC10-DET and NEU-DET validation sets.

Dataset	Model	mAP@0.5 (%)	mAP@0.5:0.95 (%)
GC10-DET	YOLO26n (Baseline)	70.86 ± 0.63	37.91 ± 0.48
	OSDM	73.99 ± 1.06	38.26 ± 0.52
	FOCR	71.70 ± 0.43	37.27 ± 0.39
	SCD-YOLO (Ours)	75.19 ± 1.36	38.86 ± 0.64
NEU-DET	YOLO26n (Baseline)	75.79 ± 0.43	44.50 ± 0.84
	SCD-YOLO (Ours)	76.84 ± 0.67	45.48 ± 0.40

Results are reported as mean ± standard deviation over three independent runs with random seeds 0, 1, and 2. All runs use the same training configuration except for the random seed.

The integration of FOCR alone into the baseline yields a marginal mAP@0.5 improvement (71.1% to 71.6%, +0.5 percentage points) but reduces mAP@0.5:0.95 by 0.9 percentage points (37.8% to 36.9%). This reduction is consistent with the design hypothesis underlying SCD-YOLO. FOCR is an amplification module; without prior background suppression, its bounded fractional-order-inspired mask amplifies both genuine defect gradients and residual periodic noise. The amplified noise degrades strict-IoU localization, which is precisely what the mAP@0.5:0.95 metric captures.

When OSDM and FOCR operate together, mAP@0.5 reaches 75.3% and mAP@0.5:0.95 reaches 38.7%, exceeding all single-module configurations on both metrics. The two modules thus contribute complementary functionalities. The role of FOCR illustrates this most clearly: in isolation, FOCR reduces mAP@0.5:0.95 by 0.9 percent points, yet when added on top of OSDM, the same module instead contributes positively, raising mAP@0.5:0.95 by 0.5 percentage points over the OSDM-only configuration. This sign reversal empirically supports the cleanse-then-refine design principle introduced in Section III: OSDM first removes the periodic background components that would otherwise be amplified by FOCR, after which FOCR effectively sharpens the weak defect gradients that survive the cleansing step. The full SCD-YOLO uses fewer parameters (2.24M vs 2.38M) and fewer GFLOPs (5.1 versus 5.2) than the baseline, confirming that the spectral redesign improves accuracy while retaining a compact parameter count and nominal computational complexity.

The runtime measurements reveal a moderate practical cost of spectral processing. SCD-YOLO increases inference latency from 2.04 ± 0.35 to 2.27 ± 0.25 ms/image relative to the baseline, while peak GPU memory increases from 299.87 to 315.77 MB. Thus, although SCD-YOLO retains fewer parameters and slightly lower nominal GFLOPs than YOLO26n, FFT-based processing introduces measurable runtime overhead that is not fully reflected by theoretical complexity.

To examine whether the reported improvements depend on a particular random initialization, we repeat the core experiments using three random seeds (0, 1, and 2), as summarized in Table 7. On GC10-DET, SCD-YOLO achieves 75.19 ± 1.36% mAP@0.5 and 38.86 ± 0.64% mAP@0.5:0.95, compared with 70.86 ± 0.63% and 37.91 ± 0.48% for YOLO26n, corresponding to average improvements of 4.33 and 0.95 percentage points, respectively. The multi-seed ablation results also preserve the qualitative trend observed in Table 6: OSDM provides a clear improvement, FOCR alone yields a modest gain in mAP@0.5 but a reduction in mAP@0.5:0.95, whereas their combination achieves the highest mean performance on both metrics. On NEU-DET, SCD-YOLO improves the mean mAP@0.5 from 75.79 ± 0.43% to 76.84 ± 0.67% and mAP@0.5:0.95 from 44.50 ± 0.84% to 45.48 ± 0.40%. These results indicate that the average improvements and the main ablation conclusions are retained across repeated training runs rather than being inferred from a single run.

Statistical testing further supports the multi-seed comparison between YOLO26n and SCD-YOLO. On GC10-DET, the 95% confidence intervals for the paired differences are [2.41, 6.25] percentage points for mAP@0.5 and [0.57, 1.33] percentage points for mAP@0.5:0.95, with paired t-test p-values of 0.0105 and 0.0084, respectively. On NEU-DET, the corresponding intervals are [0.43, 1.67] and [−0.15, 2.11] percentage points, with p-values of 0.0182 and 0.0644. The exact two-sided Wilcoxon signed-rank test returns $p = 0.250$ in all four comparisons.

E. HYPERPARAMETER SENSITIVITY ANALYSIS

1) SENSITIVITY TO THE OSDM TOP-K PARAMETER

The Top-k parameter in OSDM determines the number of dominant high-amplitude spectral positions selected as background candidates. To examine its influence, we evaluate $k \in 1, 3, 5, 7$ while keeping the OSDM placement, network architecture, and training settings unchanged. All experiments are conducted on the GC10-DET validation set.

As shown in Table 8, $k = 3$ achieves the best performance, reaching 74.1% mAP@0.5 and 38.2% mAP@0.5:0.95. When $k = 1$, only a limited number of dominant spectral responses are selected, which may leave part of the periodic background insufficiently attenuated. Increasing k to 3 improves both metrics, indicating a better balance between background suppression and defect-information preservation. Further increasing k to 5 or 7 does not provide additional gains

TABLE 8. Sensitivity analysis of the OSDM top-k parameter on GC10-DET validation set (without FOCR).

k	mAP@0.5 (%)	mAP@0.5:0.95 (%)	Params (M)	GFLOPs
1	73.5	37.6	2.16	5.0
3	74.1	38.2	2.16	5.0
5	72.8	37.4	2.16	5.0
7	73.7	37.7	2.16	5.0

TABLE 9. Sensitivity analysis of the FOCR fractional order on GC10-DET validation set.

Model	Order / Range	Initial α	Learned α_{P4}	Learned α_{P5}	mAP@0.5 (%)	mAP@0.5:0.95 (%)
Fixed	$\alpha = 0.1$	0.1	-	-	70.5	38.4
Fixed	$\alpha = 0.3$	0.3	-	-	69.6	37.1
Fixed	$\alpha = 0.5$	0.5	-	-	71.5	37.4
Learnable	[0.2,0.4]	0.35	0.2886	0.200†	71.7	37.8
Learnable	[0.1,0.5]	0.35	0.3035	0.100†	75.3	38.7
Learnable	[0.05,0.6]	0.35	0.3024	0.050†	74.7	38.5

† The learned effective order reaches the lower bound of the specified range.

TABLE 10. Ablation Study of OSDM Placement on GC10-DET (without FOCR) validation set.

Placement	mAP@0.5 (%)	mAP@0.5:0.95 (%)	Params (M)	GFLOPs
YOLO26n (Baseline)	71.1	37.8	2.38	5.2
OSDM @ P3	72.6	38.0	2.38	5.2
OSDM @ P4	69.3	37.6	2.38	5.2
OSDM @ P5 (pre-SPPF)	70.0	37.1	2.39	5.2
OSDM @ P5 (post-C2PSA)	69.2	37.9	2.39	5.2
OSDM (replace C2PSA)	74.1	38.2	2.16	5.0

TABLE 11. Ablation study of FOCR Placement on GC10-DET (with OSDM) validation set.

Placement	mAP@0.5 (%)	mAP@0.5:0.95 (%)	Params (M)	GFLOPs
YOLO26n (Baseline)	71.1	37.8	2.38	5.2
FOCR @ P3	71.0	37.3	2.17	5.1
FOCR @ P4	72.6	37.9	2.18	5.1
FOCR @ P5	74.0	38.6	2.23	5.1
FOCR @ P3+P4+P5	70.2	38.3	2.25	5.2
FOCR @ P4+P5 (SCD-YOLO)	75.3	38.7	2.24	5.1

and may increase the likelihood of including defect-relevant spectral components in the suppression set. We use $k = 3$ in the final configuration. The same setting is retained for NEU-DET without dataset-specific retuning. However, because k is selected on the GC10-DET validation set, it should be regarded as an empirically chosen setting rather than a universally optimal value; datasets with substantially different spatial resolutions, texture periodicities, or defect distributions may favor a different suppression budget.

2) FOCR FRACTIONAL ORDER

To examine the effect of the fractional-order setting in FOCR, we evaluate three fixed orders, $\alpha \in 0.1, 0.3, 0.5$, and three learnable ranges, [0.2,0.4], [0.1,0.5], and [0.05,0.6]. For all learnable configurations, the order is initialized to 0.35. The two FOCR modules at P4 and P5 learn their order parameters independently. All other architectural components and training settings are kept unchanged, and the experiments are conducted on the GC10-DET validation set. The results are summarized in Table 9.

It evaluates the sensitivity of FOCR to both fixed fractional orders and learnable order ranges. The fixed-order configurations show noticeable performance variation, indicating that a manually specified order does not consisten-tly provide the most suitable spectral response. In the learnable configurations, the P4 order converges consistently to approximately 0.29–0.31 across all tested ranges. In contrast, the effective P5 order reaches the lower bound in all three cases, suggesting that the deeper P5 branch favors a weaker fractional-order response. The narrow range [0.2,0.4] restricts P5 to $\alpha = 0.2$ and yields only 71.7% mAP@0.5 and 37.8% mAP@0.5:0.95. Relaxing the lower bound substantially improves performance. The original range [0.1,0.5]

achieves the best overall result of 75.3% mAP@0.5 and 38.7% mAP@0.5:0.95, while further widening the range to [0.05,0.6] provides no additional gain. We retain [0.1, 0.5] as the default range in FOCR, as it provides the best overall performance among the tested settings.

F. MODULE PLACEMENT ANALYSIS

1) PLACEMENT OF OSDM

To find where OSDM works best, we test five placements along the backbone. Four of them add OSDM as an extra layer after the P3, P4, pre-SPPF P5, and post-C2PSA P5 stages; the fifth replaces the C2PSA block at the P5 tail. Table 10 reports the results. The replacement strategy gives the best accuracy on both metrics, 74.1% mAP@0.5 and 38.2% mAP@0.5:0.95, with the smallest parameter count. The placement results also expose an important limitation of frequency-based cleansing. OSDM at P3 still provides a modest improvement over the baseline, whereas the P4 and the two additive P5 placements fall below the baseline in mAP@0.5. At earlier or intermediate stages, the features retain substantial low-level texture together with defect structures, and their spectral responses may overlap more strongly. Under such conditions, treating dominant spectral components as background candidates can also attenuate defect-relevant responses. OSDM is not expected to be uniformly beneficial at all feature levels; its effectiveness depends on sufficient statistical spectral separability, which is better established after semantic abstraction at the backbone tail. OSDM does not explicitly require strict periodicity because the candidate positions are selected from the amplitude distribution of each input; however, its benefit may decrease when the background spectrum is diffuse or when defect-relevant frequencies overlap substantially with dominant background components.

Replacement also proves more effective than addition. Adding OSDM after the C2PSA block reaches only 69.2% mAP@0.5 at 2.39M parameters, while replacing C2PSA raises accuracy to 74.1% and cuts the parameter count to 2.16M. C2PSA captures long-range spatial dependencies but has no mechanism to tell periodic background textures apart from defect responses; OSDM handles exactly this separation in the frequency domain. The largest gains over baseline configurations concentrate on the hardest weak-edge categories, consistent with the design intent of frequency-domain background suppression. We adopt the C-2PSA-replacement placement throughout the remaining experiments.

2) PLACEMENT OF FOCR

With OSDM fixed at the backbone tail, we next determine at which neck detection scales FOCR should operate. The neck produces three detection branches (P3, P4, P5) at strides 8, 16, and 32. We evaluate FOCR at each individual scale (P3, P4, P5), at the two deeper scales (P4 + P5), and at all three (P3 + P4 + P5). Results are reported in Table 11.

Among single-scale placements, from 71.0% mAP@0.5 at P3 to 72.6% at P4 and 74.0% at P5. This trend follows the distance from OSDM. The P5 branch lies closest to the cleansed backbone output, so FOCR enhances features whose background has already been suppressed, while the shallower branches still carry residual texture that FOCR amplifies along with the defect signal. Applying FOCR at both deeper scales (P4 + P5) gives the best result on both metrics, 75.3% mAP@0.5 and 38.7% mAP@0.5:0.95, as the P4 and P5 detection heads each receive sharper gradients. We adopt it in the final SCD-YOLO.

G. VISUALIZATION AND QUALITATIVE ANALYSIS

This section presents qualitative analyses from four complementary perspectives: side-by-side detection results on representative test samples, the spectral behaviors of OSDM and FOCR, feature-response analysis across the four ablation configurations, and representative error and failure cases.

1) QUALITATIVE DETECTION RESULTS

Figure 6 presents side-by-side inference results of YOLO26n and SCD-YOLO on four representative GC10-DET test samples under identical inference settings. The four cases illustrate complementary detection challenges. In (a), YOLO26n localizes the defect region but misclassifies the ground-truth crease as weldline, whereas SCD-YOLO correctly predicts crease. In (b), the baseline produces no detection above the common confidence threshold, while SCD-YOLO recovers the missed crease. In the multi-defect sample in (c), YOLO26n detects only the weldline, whereas SCD-YOLO additionally recovers the crescent_gap defect. Finally, in (d), both models correctly identify the waist_folding defect, but SCD-YOLO improves the bounding-box IoU from 0.68 to 0.78. These examples illustrate improvements in class discrimination, missed-defect recovery, multi-defect recall, and localization quality.

2) SPECTRAL BEHAVIOR OF OSDM

Figure 7 visualizes how OSDM operates in the frequency domain on a representative sample. Subfigure (a) shows the input image, where periodic rolling marks are visible across the steel substrate. Subfigure (b) shows the amplitude spectrum of the OSDM input feature map on a log scale. The energy concentrates near the spectral center and along the horizontal and vertical axes, a pattern that reflects large-scale background structure rather than localized defect signals.

Subfigure (c) shows the effective suppression factor $1 - \alpha\mathcal{B}$, where α is the sample-adaptive suppression strength and $\mathcal{B}$ is the binary mask selecting the top-k highest-amplitude frequencies. Darker regions indicate stronger attenuation. The mask concentrates suppression on the highest-energy regions, namely the spectral center and the horizontal and vertical bands visible in (b). Subfigure (d) shows the OSDM output spectrum. The high-energy components are clearly reduced relative to (b), while the off-axis

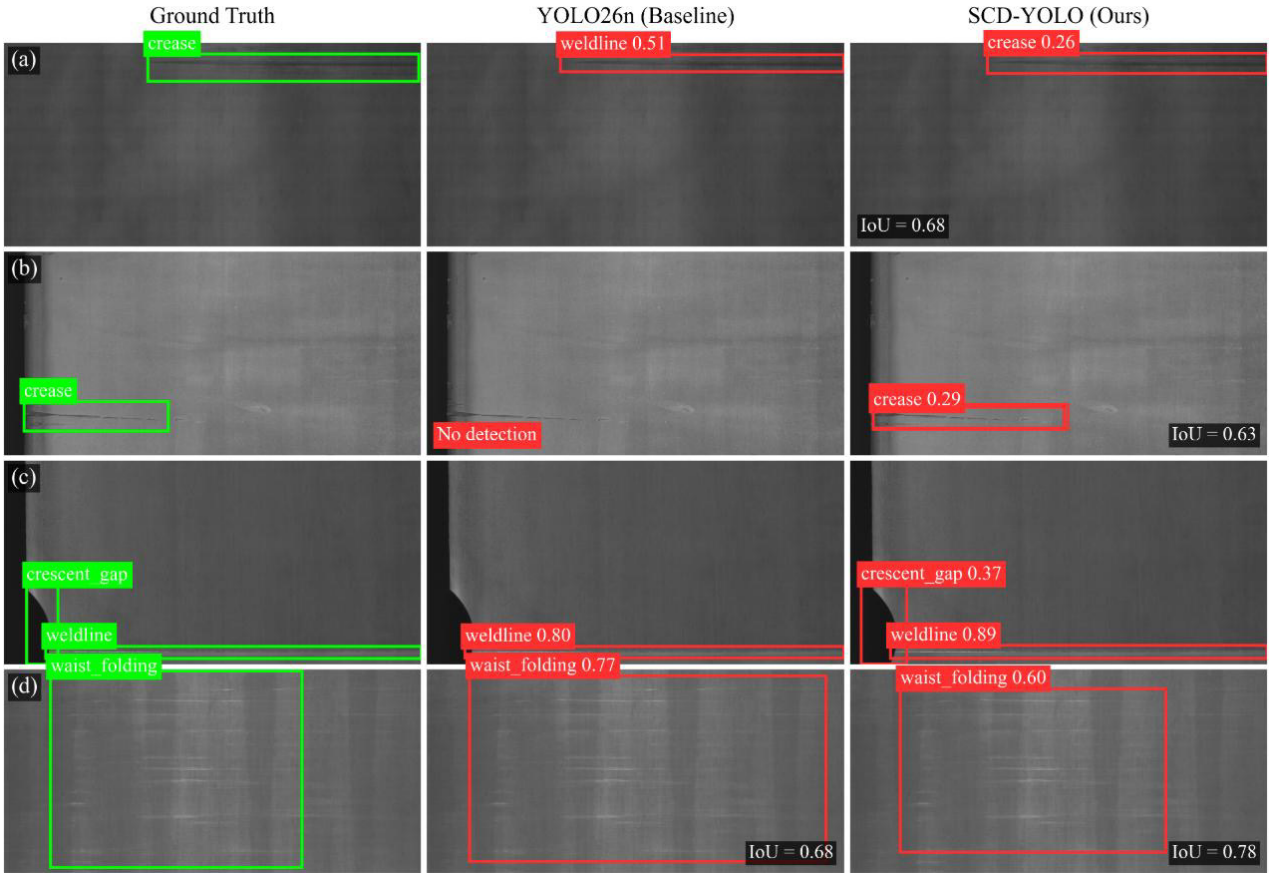


FIGURE 6. Qualitative comparison of YOLO26n and SCD-YOLO on representative GC10-DET test samples.

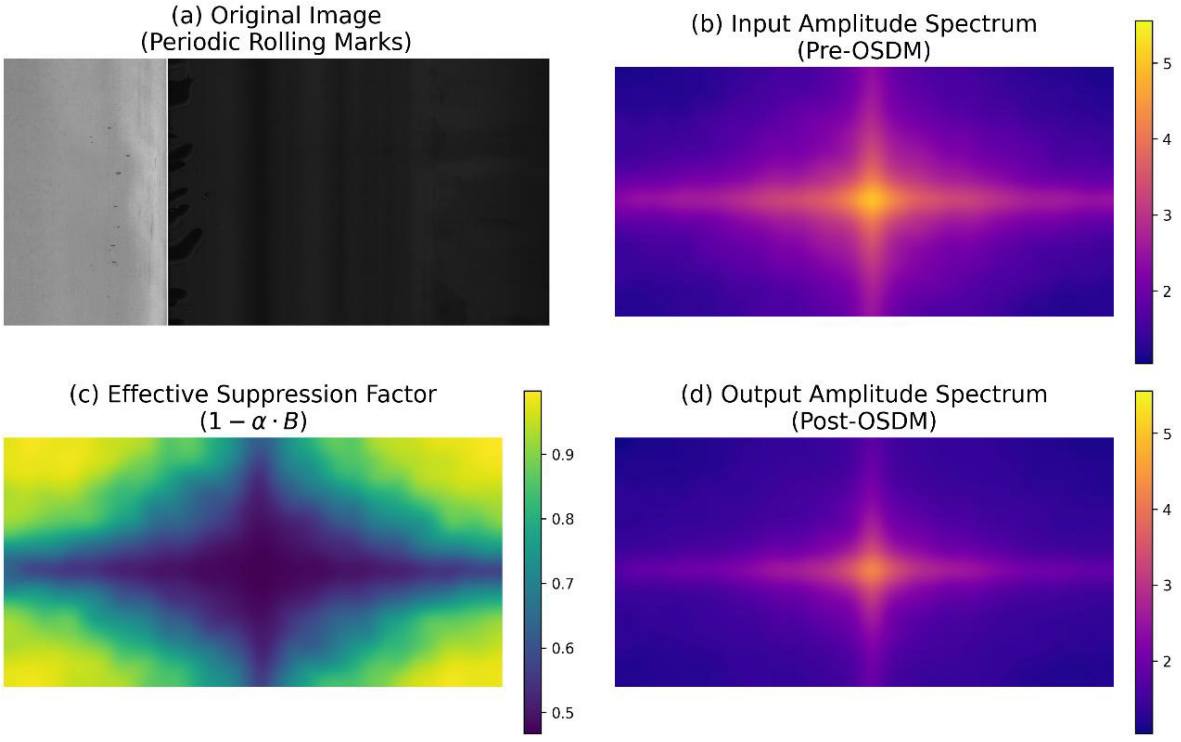


FIGURE 7. Spectral behavior of OSDM.

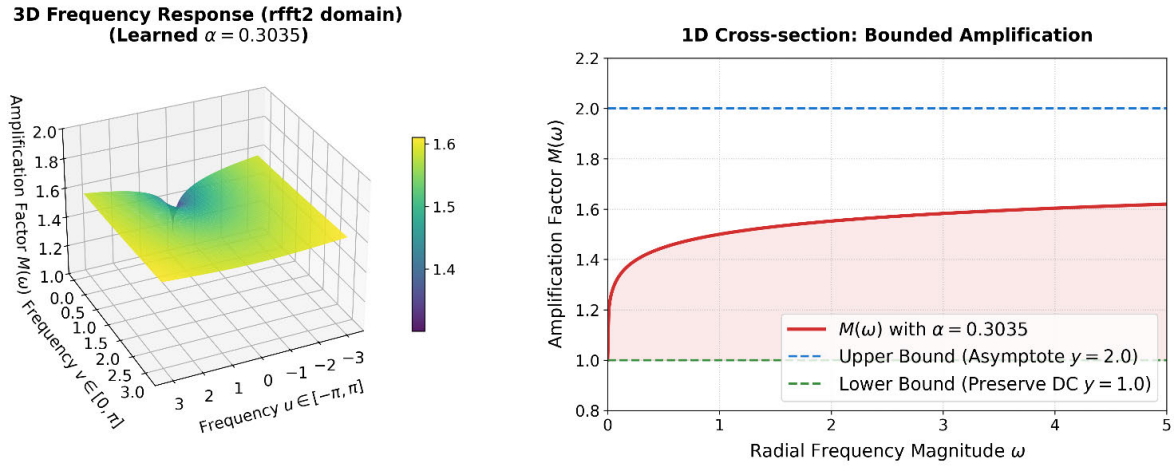


FIGURE 8. FOCR frequency response m(ω).

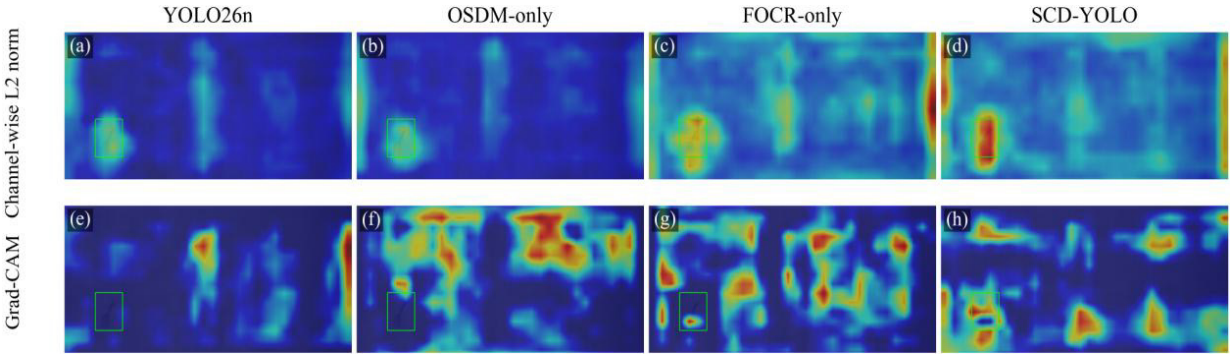


FIGURE 9. Feature-response and Grad-CAM visualizations across the four model configurations on the same representative GC10-DET test sample.

frequencies are largely preserved. This behavior matches the OSDM design in Section III. The selected components are therefore better interpreted as dominant spectral candidates rather than exclusively background frequencies. OSDM softly attenuates these responses while preserving phase, reducing dominant texture-related interference before downstream processing.

3) FOCR FREQUENCY-RESPONSE ANALYSIS

Figure 8 visualizes the frequency response of the P4 FOCR, $m(\omega) = 1 + \omega^{\alpha} \div (1 + \omega^{\alpha})$, evaluated at its learned effective order $\alpha_{P4} = 0.3035$ in the trained SCD-YOLO model. The 3D surface shows the response over the rfft2 frequency plane. It reaches its lowest value at the DC component, preserving the image mean, and rises smoothly toward higher frequencies. The 1D radial cross-section confirms that the amplification factor stays bounded between 1 at DC and 2 in the high-frequency limit, as derived in Section III. The P4 branch converges to an effective order of approximately 0.30, producing a moderate bounded spectral response. As shown by the sensitivity analysis in Table 9, the independently learned P5 order reaches the lower bound of the specified range, confirming that the two feature scales favor different spectral responses.

4) FEATURE-RESPONSE AND GRAD-CAM ANALYSIS ACROSS CONFIGURATIONS

Figure 9 compares the P5 feature responses and Grad-CA-M attribution maps of YOLO26n, OSDM-only, FOCR-only, and SCD-YOLO on the same GC10-DET test sample. Figures 9(a)–(d) show channel-wise L2 responses with shared normalization, whereas Figures 9(e)–(h) show Grad-CAM maps for the target defect class. The Grad-CAM target is the strongest target-class logit within the ground-truth region at the P5 detection scale.

In the L2 maps, YOLO26n and OSDM-only exhibit relatively moderate P5 responses, whereas FOCR-only increases the response around the defect while also producing elevated responses over surrounding structures. This behavior is consistent with the refinement role of FOCR when it operates without prior spectral cleansing. The full SCD-YOLO produces a pronounced feature response within the ground-truth region, although elevated responses remain elsewhere in the P5 representation.

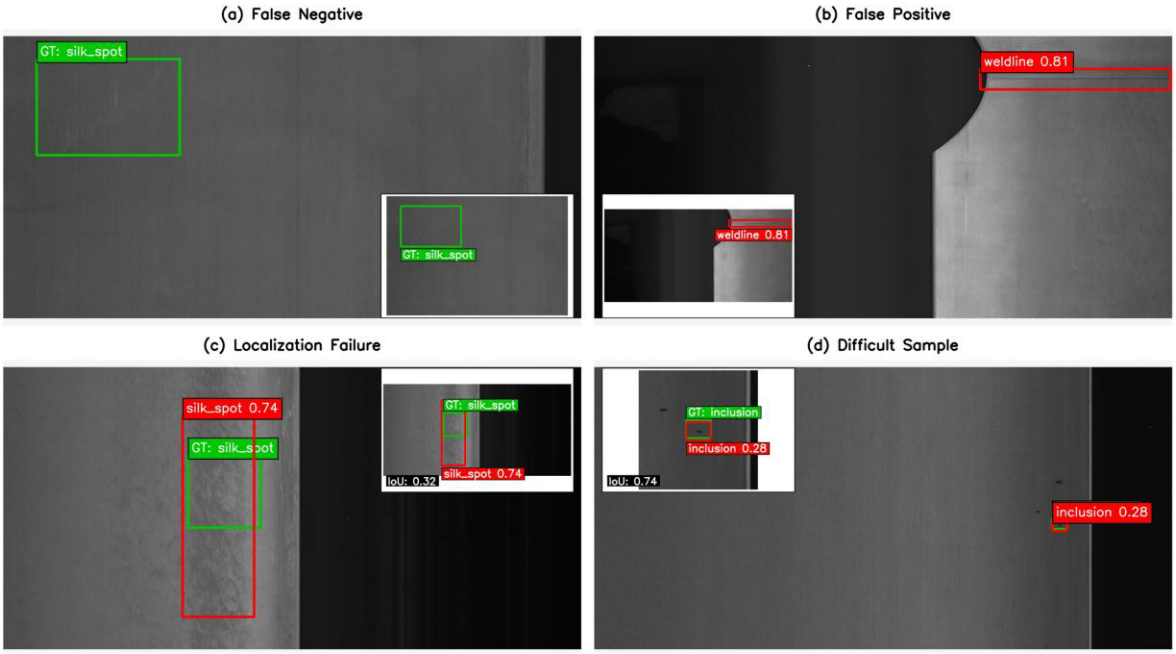


FIGURE 10. Representative error and difficult cases of SCD-YOLO on the GC10-DET test set. (a) False negative for a low-contrast silk-spot defect. (b) False positive associated with complex background structure. (c) Localization failure in which the defect class is correctly identified but the predicted bounding-box extent is inaccurate. (d) A difficult small and low-contrast inclusion that is correctly detected with low confidence. Green and red boxes denote ground-truth and predicted bounding boxes, respectively. Enlarged regions are shown for visual clarity, with IoU reported for the localization-related cases.

The Grad-CAM maps show that the baseline and single-module configurations retain substantial attribution outside the defect region, whereas SCD-YOLO places its dominant response within the ground-truth box, despite some residual background attribution. This pattern is consistent with the complementary effects of spectral cleansing and subsequent refinement.

5) ERROR AND FAILURE-CASE ANALYSIS

A complementary error and difficult-case analysis is conducted using the final SCD-YOLO model on the GC10-DET test set. A false negative is defined as a ground-truth defect without a matched prediction at IoU ≥ 0.5, a false positive as an unmatched prediction, and a localization failure as a correctly classified prediction with IoU in the range $[0.2, 0.5)$. Difficult samples are additionally identified according to weak confidence, small defect area, and low local contrast. Figure 10 presents representative examples.

The results show that missed detections are mainly associated with weak or diffuse defects, false positives are often induced by complex background structures, and localization failures occur when the defect extent is ambiguous. These cases indicate that, although SCD-YOLO improves overall detection accuracy, extremely weak defect signals and complex textures remain challenging.

H. LIMITATIONS AND FUTURE WORK

Although SCD-YOLO improves detection performance on both GC10-DET and NEU-DET, several limitations remain. First, the current evaluation is restricted to two steel-surface defect datasets, and the effectiveness of the proposed spectral modules on non-metallic surfaces or backgrounds with weak periodicity has not yet been established. In addition, GC10-DET and NEU-DET are trained and evaluated independently because their defect label spaces are not directly aligned; therefore, the present experiments do not constitute direct cross-domain or zero-shot transfer evaluation.

The failure-case analysis also shows that extremely weak or low-contrast defects, complex background structures, and ambiguous defect boundaries remain challenging, leading to missed detections, false positives, or inaccurate localization in some cases. Furthermore, the current runtime evaluation is conducted only on an RTX 4090, so deployment efficiency on resource-constrained industrial or edge hardware has not yet been validated.

Future work will evaluate SCD-YOLO on embedded platforms, investigate more robust spectral adaptation for difficult defect-background overlap, and extend the framework to other inspection domains, such as fabric and PCB defect detection.

V. CONCLUSION

This paper proposed SCD-YOLO, a frequency-domain extension of YOLO26n for industrial surface defect detection. The method introduces two complementary modules: OSDM attenuates dominant background frequency components at the backbone tail while preserving phase, and FOCR performs bounded spectral refinement at the P4 and P5 neck branches to enhance weak defect-relevant responses. Together, they form a cleanse-then-refine framework

in which background interference is first suppressed and the remaining informative responses are subsequently refined.

Final evaluations on the held-out GC10-DET and NEU-DET test sets demonstrate that SCD-YOLO consistently improves the YOLO26n baseline while retaining low model complexity. The validation-based ablation and placement studies further support the proposed design: OSDM is most effective at the backbone tail, whereas FOCR performs best on the deeper neck branches receiving the cleansed representation. The combined model consistently outperforms the single-module variants, indicating that prior spectral cleansing is important before subsequent refinement. The relative gains under loose and strict IoU criteria, however, vary across datasets. SCD-YOLO retains a compact parameter count and nominal computational complexity primarily because OSDM replaces C2PSA and FOCR is restricted to the deeper P4 and P5 scales rather than being applied throughout the network.

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